torch.inner#

https://pytorch.org/docs/stable/generated/torch.inner.html

Description:

Computes the dot product for 1D tensors. For higher dimensions, sums the product of elements from input and other along their last dimension. If either input or other is a scalar, the result is equivalent to torch.mul(input, other). If both input and other are non-scalars, the size of their last dimension must match and the result is equivalent to torch.tensordot(input, other, dims=([-1], [-1])) input (Tensor) – First input tensor

Function Signature

```
torch.inner(input, other, *, out=None) → Tensor#
Parameters
Keyword Arguments
```

Parameters

Keyword

out (Tensor, optional) – Optional output tensor to write result into. The output shape is input.shape[:-1] + other.shape[:-1].

Code Examples

```
# Dot product
>>> torch.inner(torch.tensor([1, 2, 3]), torch.tensor([0, 2,
1]))
tensor(7)

# Multidimensional input tensors
>>> a = torch.randn(2, 3)
>>> a
tensor([[0.8173, 1.0874, 1.1784],
        [0.3279, 0.1234, 2.7894]])
>>> b = torch.randn(2, 4, 3)
>>> b
tensor([[[-0.4682, -0.7159,  0.1506],
        [ 0.4034, -0.3657,  1.0387],
        [ 0.9892, -0.6684,  0.1774],
        [ 0.9482,  1.3261,  0.3917]],

        [[ 0.4537,  0.7493,  1.1724],
        [ 0.2291,  0.5749, -0.2267],
        [-0.7920,  0.3607, -0.3701],
        [ 1.3666, -0.5850, -1.7242]]])
>>> torch.inner(a, b)
tensor([[[-0.9837,  1.1560,  0.2907,  2.6785],
        [ 2.5671,  0.5452, -0.6912, -1.5509]],

        [[ 0.1782,  2.9843,  0.7366,  1.5672],
        [ 3.5115, -0.4864, -1.2476, -4.4337]]])

# Scalar input
>>> torch.inner(a, torch.tensor(2))
tensor([[1.6347, 2.1748, 2.3567],
        [0.6558, 0.2469, 5.5787]])
```

Notes & Warnings

NOTE Note If either input or other is a scalar, the result is equivalent to torch.mul(input, other). If both input and other are non-scalars, the size of their last dimension must match and the result is equivalent to torch.tensordot(input, other, dims=([-1], [-1]))

Detailed Documentation

Documentation

Computes the dot product for 1D tensors. For higher dimensions, sums the product of elements from input and other along their last dimension.

If either input or other is a scalar, the result is equivalent to torch.mul(input, other).

If both input and other are non-scalars, the size of their last dimension must match and the result is equivalent to torch.tensordot(input, other, dims=([-1], [-1]))

input (Tensor) – First input tensor

other (Tensor) – Second input tensor

out (Tensor, optional) – Optional output tensor to write result into. The output shape is input.shape[:-1] + other.shape[:-1].